

Upper Rhine Robotic Mobile Challenge

2025

Contest Rules

September 2025

Introduction

The Upper Rhine Robotic Mobile Challenge (UR2MC) is a competition open to Bachelor's and Master's level students from technological programs across Europe. It has three main objectives. First, preparing for the challenge serves as a motivating factor for students, allowing them to practically apply the technical knowledge and skills they have learned, and it can also support their academic training. Second, by promoting a positive image, this event aims to highlight scientific and technological programs dedicated to robotics during the BE 5.0 trade show. Lastly, the competition provides a valuable opportunity for interaction and camaraderie between students and industry professionals, fostering strong ties between academia and businesses.

This contest rules give the conditions for participating with the following objectives :

- Allow participation at minimal financial cost
- Showcase with high-performing robots.
- Demonstrate the expertise of academic programs.
- Promote fair play during the event.
- Simplify the jury arbitration.
- Ensure fairness among the different teams.

1 First rule- rules limits – Modification

The organizing committee reserves the right to modify these rules at any point during the competition if it deems it necessary to uphold the spirit of the event. In such cases, a meeting with the remaining teams will be organized to inform all participants, and the changes will be posted.

Sovereign Jury : The jury is sovereign, and its decisions are final. It may choose to penalize a robot and/or a team that exhibits behavior contrary to the spirit of the competition, even if the infraction is not explicitly covered by these rules. There is no video review. Any objection based on video footage will not be considered.

2 Second rule – behavior during the event

2.1 Students behavior

The students represent their university/school/college in a public event. As such, a professional attitude is expected : a clean and organized booth, and respectful behavior. Alcohol is prohibited at the event. It is also important to respect the venue and maintain the same behavior outside the event. Any violation of this rule will result in the immediate expulsion of the students involved.

2.2 Tutor behavior

This is a student competition. Tutors accompanying a team are **STRICTLY FORBIDDEN** from being at their team's booth to assist in the setup or adjustment of the robots. The proper role of a tutor is to serve as part of the jury. However, they may assist students from their team or another team by suggesting improvements, but under no circumstances are they allowed to directly work on the robot or computers.

3 Third rule - Robot description

The robot is a Turtlebot3 burger developed by ROBOTIS¹(see Figure 1).

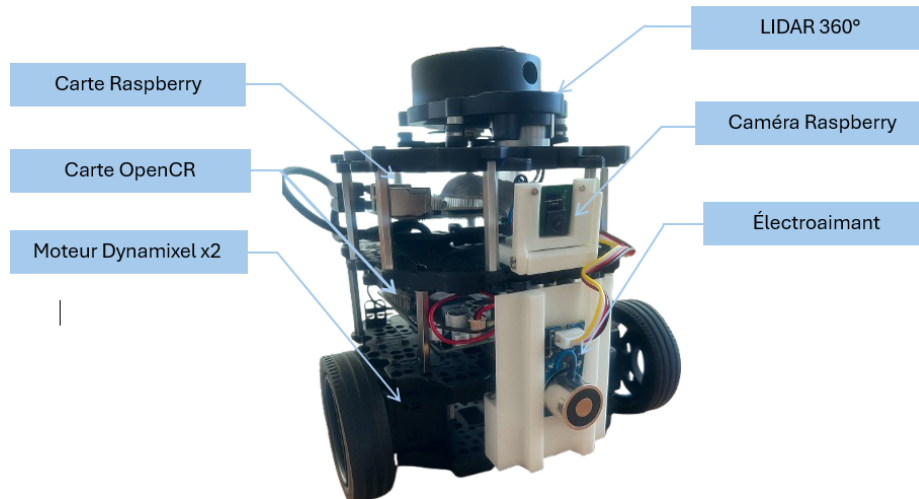


FIGURE 1 – Turtlebot3 burger robot

It measures 138 x 178 x 192 mm and weighs approximately 1 kg. Its maximum linear speed is 0.22 m/s, and its maximum rotational speed is 162.72°/s. In terms of perception, it is equipped with a 360° LIDAR, a Raspberry Pi color camera mounted on a bracket with adjustable lateral angle, and an IMU. The control system is managed by a Raspberry Pi 4 or 3 and an OpenCR board powered by an STM32 processor. Two DYNAMIXEL motors ensure its mobility (see Figure 1), and a Li-Po battery provides energy autonomy of approximately 2.5 hours. As part of this competition, the robot

1. <https://emanual.robotis.com/docs/en/platform/turtlebot3/overview/>

is also equipped with a GROVE Electromagnet module, capable of generating a magnetic field when powered, allowing it to attract and hold small ferromagnetic metal objects.

Software development framework is ROS (Robot Operating System) or ROS2. The Single Board Computer (SBC) Raspberry embedded a Linux Ubuntu operating system with a serial communication to OpenCR board. Programming languages are Python, C++ or MATLAB. Robot has to be autonomous but a part of the control can be decentralized.

The robots are all identical and will remain so throughout the event. It is prohibited to modify their structure or add new equipment. Violating this rule will result in the immediate disqualification of the team. The ROS middleware is mandatory but development language is free.

4 Fourth rule - Robot missions

All of the missions are related to issues encountered in the field of logistics with AMR-type robots (Autonomous Mobile Robot). Robots have to drive on a track with a width around 20 cm. The track is in wood and bounded by two yellow and white lines with a width around 2 cm (voir Figure 2). The track is on a 4m x 3m platform made of wood delimited by two bands of white and yellow color of 2 cm width. The track is set on a 4m x 3m platform covered with a black carpeting.

The track consists of curves, straight lines, obstacles area, parking area, a closed area. The missions are described below.

4.1 Track following

As part of a logistics mission, the robot may need to move from one point to another along a specific path, represented here by a track. This mission is split in two parts :

1. First, follow a track consisting of a straight line followed by a curve, over a length of approximately 1.5 meters. The track is marked along by a white line on the right and a yellow line on the left. Stop at the crossing of a line marked by a black and white checkered pattern, which is perpendicular to the track. The robot is placed on a starting line marked by a white circle centered on the track. This task is used for the qualification phase (see the challenge organization).
2. Follow a track consisting of a straight line followed by a curve but with yellow or white line disrupted. The disruption is around 50 cm. The robot has to stop when it detects a STOP sign with the 5×5 cm size.

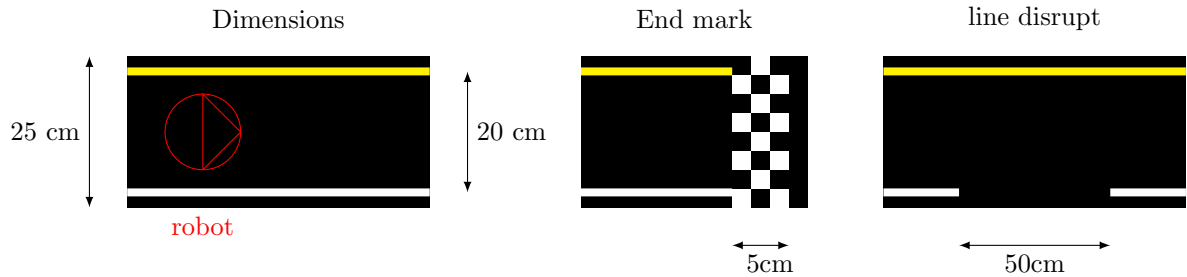


FIGURE 2 – Track dimension and others

4.2 Obstacle avoiding

In a warehouse, obstacles can impede traffic. The robot must be able to avoid them. On an area bounded by white and yellow lines, the robot has to avoid obstacles without to drive outside the area and at the end stop in front of a barrier closing the road (see Figure 3).

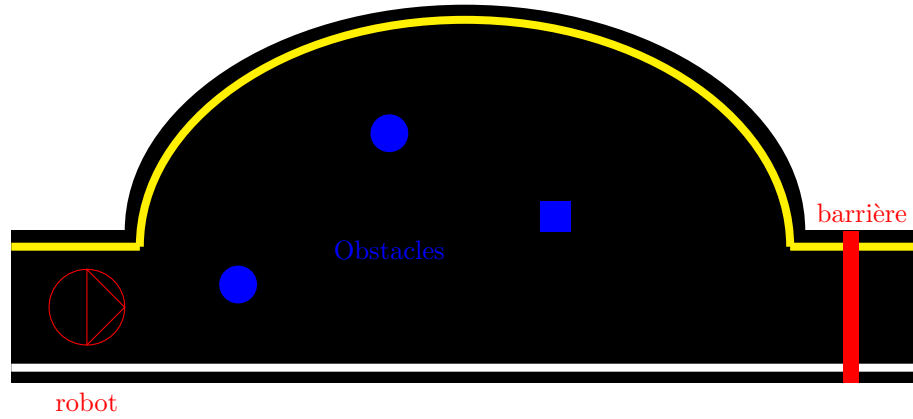


FIGURE 3 – Obstacles area.

4.3 Driving in a closed area

The warehouse may contain certain areas where visibility is severely impaired and, as a result, the camera cannot be used as a perception system. This area is represented here by an enclosure bounded by walls with an entrance and an exit. (see Figure 4).

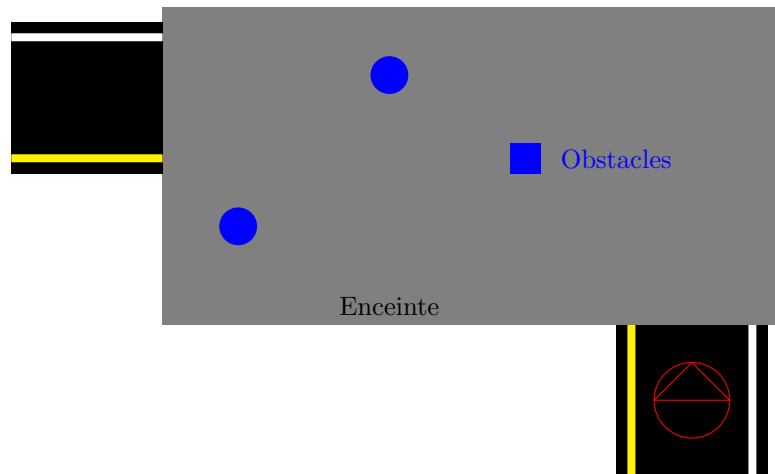


FIGURE 4 – Closed area with camera forbidden.

The mission is split in 3 parts :

1. Robot has to detect the input and step without touch the wall.

- For this mission, camera is forbidden.

The storage/retrieval area consists of three separate zones numbered ① to ③ by a sign placed at the rear of each zone but defined randomly (see figure 5. Zone ① is initially occupied by a pallet measuring 15 cm $\times$ 15 cm $\times$ 5 cm resting on a 10 cm high support. The front part of the pallet has a metal section that allows the robot to grab it using an electromagnet (Grove electromagnet type). The mission consists of moving the pallet from zone ① to zone ②. It can be broken down into four parts : This mission can be broken down into three parts :

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- The diagram illustrates a four-bay portal frame structure. It consists of four vertical columns and three horizontal beams. The columns are labeled 1, 2, 3, and 4 from left to right. The beams are labeled 1, 2, and 3 from top to bottom. The structure is supported by a fixed base. A dashed line indicates the ground level. A yellow line is shown at the bottom of the diagram.

The electromagnet control is left free. Four landmarks (fixed reference points that can be identified without ambiguity and are used for navigation) positioned at the four corners of the area can be used for localization.



FIGURE 6 – All the driving signs used for the missions

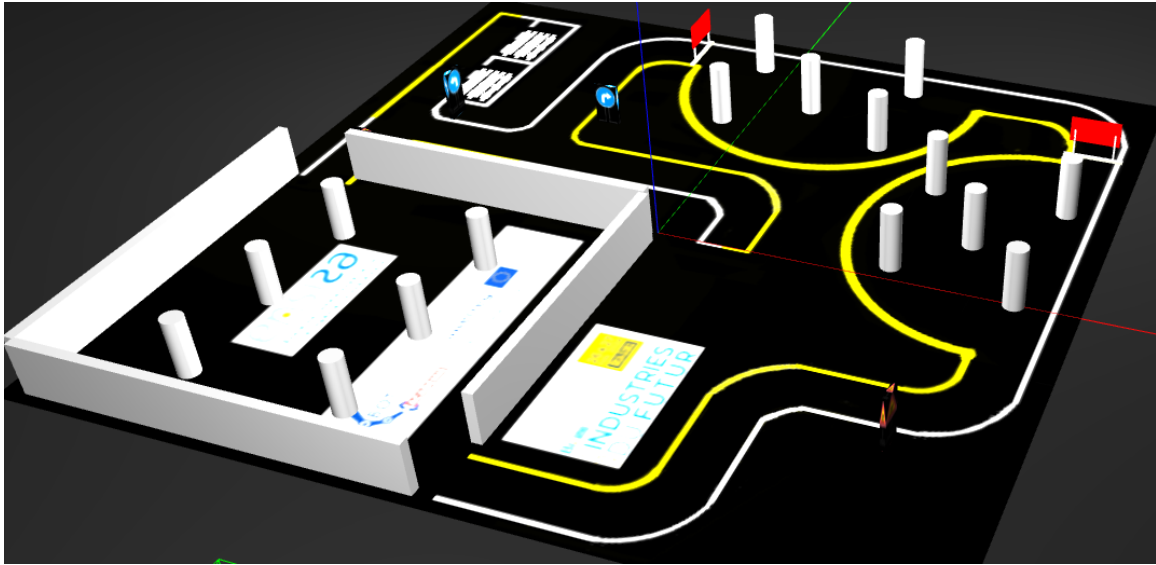


FIGURE 7 – Illustration of the robot challenge environment in Gazebo

5 Challenge organization

The challenge is organized in five steps : registration, development, qualification, challenge, and the awards ceremony.

5.1 Registration

To participate in the contest, a registration must be submitted to the organizers by the dates specified on the competition's website, including the number of teams and the names of the supervising tutors. This registration is then validated in September with the names of the 3 student team members, their email addresses, the name of the institution, and the names of the supervising tutors responsible for the team. On-site, the team must present itself to the organizers on the indicated dates and times to specify the chosen name for the robot.

5.2 Development

After the registration , students will access a simulation environment (see figure 7) with :

- a tracks example,
- a robot simulation model,
- missions examples.

This welcoming pack can help the student to develop their own scripts to execute the different missions. Please note that the **ROS** or **ROS2** environment must be used. Participants are free to choose their programming language.

5.3 Qualification

The competition takes place over the 2 days of the BE 5.0 trade show. The teams are welcomed during the first half-day. A robot is then provided to the teams along with the protocol for connecting to a local network (if necessary). Qualification involves completing the first point of the first task. Once this step is validated and timed by the jury, it grants access to the completion of other tasks.

However, if a team wishes to test other tasks before having completed the track-following task within the allotted time (15 minutes), they may do so, but it will be outside of the competition.

5.4 Challenge

Missions have to follow the planning below :

- Day 1 : 14h à 17h30 – Mission 1 complete + Mission 2
- Day 2 : 9h – 13h00 – Mission 3 and 4
- Day 2 : 14h00 – 15h30 – Secret mission

To test their applications, 15-minute time slots will be made available to the teams, allowing them to access the track and validate their results. Portions of the track will also be available for testing at the booths. However, depending on track availability, tasks may be tested in a different order upon request from the teams to the jury.

5.5 Points attribution

To establish a ranking, points are awarded according to the distribution outlined in the table 1

5.6 Awards ceremony

The challenge is closed by the awards ceremony for the first four teams. It is planned on the 2^{ème} day from 15h30 with all the teams and officials.

No Mission	Definition	Points	Bonus	Malus
1	Track following	5		-1 (if out)
	Stop on mark		+1	
	Line disrupt	3		
	Stop on the STOP sign		+1	
2	Obstacles avoiding	5		-1 if one obstacle falls down, exit after 3.
	Stop at the barrier		+1	
3	Step in the closed area	3		-1 (if a wall is touched)
	Step out the area	3		-1 (if a wall is touched)
	With obstacle		+3	-1 (if one obstacle falls down)
4	Storage maneuver + pallet detection	3		-1 (if no stop)
	Pallet pickup	1		
	Pallet transport	3		
	Pallet drop-off	1	+1 if leaving the zone	
5	Secret mission	10		-1 per final position (Timing)

TABLE 1 – Points attribution.

Article 6 - Photo right

As part of the competition, photos and videos are produced and processed. Their purpose is to document the activities of the competition. The collected data is electronically saved and processed. For example, it may be published on the organizers' website, the participating institutions' websites, their social media, flyers, posters, etc. Some photos or videos may also be shared with the INTERREG Program managing authority, sponsors, or cooperation partners solely for the purpose of documenting the competition.



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